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Programmable pill dispensing device and methods of use

Abstract

A programmable pill dispensing device, system, and methods are provided. A programmable pill dispensing device may be used to automatically dispense a pill, such as a prescription medication, at a programmed predetermined time.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of Non-provisional application Ser. No. 17/882,801 filed on Aug. 8, 2022, and entitled “A PROGRAMMABLE PILL DISPENSING DEVICE AND METHODS OF USE,” which is a continuation of Non-provisional application Ser. No. 17/506,980 filed on Oct. 21, 2021, and entitled “PROGRAMMABLE PILL DISPENSING DEVICE AND METHODS OF USE,” the entirety of which are all incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention generally relates to the field of secured and automated dispensing. In particular, the present invention is directed to a programmable pill dispensing device.

BACKGROUND

(2) Often consumers do not properly take prescription medications. Consumers make take the incorrect amount of a medication or forget to take a medication, which may be harmful or even lethal.

SUMMARY OF THE DISCLOSURE

(3) In an aspect, a programmable pill dispensing device is provided, wherein the device includes a housing, which includes: a channel having a first section and a second section, a storage cavity connected to the first section of the channel, wherein the storage cavity is configured to store a plurality of pills, at least a sensor, wherein the at least a sensor is configured to detect at least a sensed property, and a tray connected to the second section of the channel. The device also includes a rotational dispensing element, wherein the dispensing element configured to dispense at least a pill of the plurality of pills, wherein the rotational dispensing element includes a body capable of rotating about a central axis and a structure disposed on the body, wherein the structure is configured to engage with the at least a pill of the plurality of pills. The device also includes a controller communicatively connected to the at least a sensor, wherein the controller is configured to actuate the rotational dispensing element as a function of the at least a sensed property, and wherein, upon an actuation, the rotational dispensing element is configured to move at least a pill of the plurality of pills from the first section of the channel to the second section of the channel and further into the connected tray in accordance with a rotation of the body.

(4) In another aspect, a method of automatedly dispensing a pill using a programmable pill dispensing device is provided, the method including storing a plurality of pills in a storage cavity contained within a housing of a programmable pill dispensing device; detecting at least a sensed property disposed in the channel using at least a sensor; actuating a rotational dispensing element of the device using the a controller communicatively connected to the at least a sensor; moving, upon actuation of the rotational dispensing element, the at least a pill of the plurality of pills through from a first section of a channel of the device to a second section of the channel of the device, which wherein the first section is connected to extends from the storage cavity and the second section is connected to a tray of the device; and alerting a user using an alarm transducer that the at least a pill is dispensed and ready for retrieval by the user.

(5) These and other aspects and features of non-limiting embodiments of the present invention will become apparent to those skilled in the art upon review of the following description of specific non-limiting embodiments of the invention in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For the purpose of illustrating the invention, the drawings show aspects of one or more

embodiments of the invention. However, it should be understood that the present invention is not limited to the precise arrangements and instrumentalities shown in the drawings, wherein:

- (2) FIG. 1A is a diagrammatic representation illustrating an isometric view of an exemplary embodiment of a programmable pill dispensing device in accordance with aspects of the invention;
- (3) FIG. 1B is a diagrammatic representation illustrating a cross-sectional view of the exemplary embodiment of the programmable pill dispensing device taken along line 1B-1B of FIG. 1A in accordance with aspects of the invention;
- (4) FIG. 2 is a diagrammatic representation illustrating a partially transparent isometric view of the exemplary embodiment of the pill dispensing device in accordance with aspects of the invention;
- (5) FIGS. 3A-3D are diagrammatic representations illustrating various exemplary embodiments of a pill dispensing element in accordance with aspects of the invention;
- (6) FIG. 4 is a flow diagram illustrating an exemplary method of automatically dispensing a pill using the exemplary embodiment of the programmable pill dispensing device;
- (7) FIG. 5 is a block diagram illustrating an exemplary machine-learning module that can be used to implement any one or more of the methodologies disclosed in this disclosure and any one or more portions thereof in accordance with aspects of the invention;
- (8) FIG. 6 is a schematic diagram illustrating exemplary embodiments of fuzzy sets; and
- (9) FIG. 7 is a block diagram illustrating a computing system that can be used to implement any one or more of the methodologies disclosed in this disclosure and any one or more portions thereof.

The drawings are not necessarily to scale and may be illustrated by phantom lines, diagrammatic representations and fragmentary views. In certain instances, details that are not necessary for an understanding of the embodiments or that render other details difficult to perceive may have been omitted.

DETAILED DESCRIPTION

(10) At a high level, aspects of the present disclosure are directed to a programmable pill dispensing device, system, and methods. Administration of pills, such as prescription medications, is a crucial process that requires precision and care. A programmable pill dispensing device may assist primary care provider (PCPs), caregivers, or living assistance providers (LAPs) with managing medication administration throughout the day for a patient. Furthermore, a programmable pill dispensing device may eliminate any questions of whether a patient took a medication at the required time.

(11) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, that the present invention may be practiced without these specific details. As used in this disclosure, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described in this disclosure as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description in this disclosure, the terms “top”, “bottom”, “upper”, “lower”, “front”, “rear”, “right”, “left”, “vertical”, “horizontal”, and derivatives thereof shall relate to the invention as oriented in FIG. 1A. Moreover, terms such as “a”, “an”, and “the”, and derivatives thereof shall be understood to mean “one or more” in this disclosure unless explicitly described as otherwise.

(12) Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed in this disclosure are not to be considered as limiting, unless

the claims expressly state otherwise.

(13) A “pill” may be any compact substance that may be consumed by swallowing, dissolving, chewing, or the like. For example, a pill may be a prescription medication, an over-the-counter medication, or a supplement, such as a vitamin supplement.

(14) Referring now to the drawings, FIGS. 1A and 1B illustrate an exemplary embodiment of a pill dispensing device **100** in accordance with one or more embodiments of the present disclosure. In one or more embodiments, pill dispensing device **100** (also referred to in this disclosure as a “device”) includes a housing **104**. Housing **104** may include a storage cavity **116** (also referred to in this disclosure as a “device”), which may be contained within housing **104** and configured to store pills **120**. Housing **104** may also include a channel **124**, which may be defined by housing **104** and may extend from storage cavity **116**. Housing **104** may also include a tray **128** that is configured to hold a dispensed pill. Channel **124** may be connected to an opening of tray **128**.

(15) With continued reference to FIG. 1, housing **104** may provide a protective layer of material configured to create a barrier between internal components of device **100** and environmental components. In one or more embodiments, housing **104** may be constructed of materials and/or combinations of materials that may include metallic materials like aluminum, aluminum alloys, steel alloys, copper, tin, titanium, another undisclosed material, or a combination thereof. In one or more embodiments, housing **104** include nonmetallic materials alone or in combination with metallic components permanently or temporarily connected together. Nonmetallic materials that may be used alone or in combination in the construction of housing **104** may include high density polyethylene (HDPE), polypropylene, polycarbonate, acrylonitrile butadiene styrene, polyethylene, nylon, polystyrene, polyether ether ketone, or any combination thereof. Housing **104** may be manufactured by a number of processes alone or in combination, including but limited to, machining, milling, forging, casting, 3D printing (or other additive manufacturing methods), turning, injection molding, or any combination thereof. One of ordinary skill in the art would appreciate that housing **104** may be manufactured in pieces and assembled together by screws, nails, rivets, dowels, pins, epoxy, glue, welding, crimping, or another undisclosed method alone or in combination. Housing **104** may include an injection molded component. The injection molded component may include a component manufactured by injecting a liquid into a mold and letting it solidify, taking the shape of the mold in its hardened form. In other embodiments, housing **104** may be made from fiberglass. In other embodiments, housing **104** may be made from a polymer. Such as, without limitations, housing **104** may be formed as multiple pieces or as a monolithic structure from a mold using plastic or resin. In one or more embodiments, housing **104** may include liquid crystal polymer, polypropylene, polycarbonate, acrylonitrile butadiene styrene, polyethylene, polyether ether ketone, and the like.

(16) As understood by one skilled in the art, housing **104** may have various forms. For example, though housing **104** is shown as a rectangular structure in FIGS. 1A-1B, housing **104** may include other various shapes and sizes that allow housing **104** to contain one or more storage cavities **116**. In one or more embodiments, housing **104** may be a monolithic structure. For instance, and without limitation, housing **104** may be a singular structure with cavity **116** and channel **124** integrated into housing **104**. In other embodiments, housing **104** may include various components. For example, housing **104** may include a separate surface that defines cavity **116** so that cavity **116** may be individually removed from device **100**. For instance, and without limitation, cavity **116** may be removed from device **100** so that, for example, a caregiver may refill storage cavity **116** with pills **120**. In one or more embodiments, housing **104** may be made from various materials and using various methods, such as using a pour mold or casting. In an embodiment, housing **104** may be manufactured using additive manufacturing and/or 3D printing methodologies.

(17) In one or more embodiments, housing **104** may include an access opening (not shown) with a housing panel **162** so that cavity **116** may be accessed by a user or medical professional. In one or more embodiments, an access opening may be located at the rear of housing **104**. For instance, and

without limitation, cavity **116** may be removed from device **100** through an access opening after unlocking panel **162**. In other instances, and without limitation, cavity **116** may be filled with pills **120** using refill tube **132**. For example, after opening panel **162**, cavity **116** may be refilled with pills **120**. Housing **104** may have a display **136** attached thereto, as discussed further below.

(18) Still referring to FIGS. **1A** and **1B**, in one or more embodiments, storage cavity **116** may be defined by an inner surface **140** of housing **104** (shown in FIG. **1B**). As understood by one skilled in the art, cavity **116** may include various shapes and sizes. Furthermore, housing **104** may define one or more cavities **116** therein. For instance, and without limitation, a plurality of cavities **116** may be disposed within housing **104** and positioned adjacent to each other (as shown in FIG. **2**). In one or more embodiments, storage cavity **116** and channel **124** may be integrated such that inner surface **140** is a monolithic piece of housing **104** that defines both storage cavity **116** and channel **124**, storage cavity **116** and channel **124** may be separate components so that cavity **116** may be separated from housing **104** and/or channel **124** as a container. For example, inner surface **140** and cavity **116** may be removed from device **100** so that cavity **116** may be filled with pills **120** or cleaned.

(19) Still referring to FIGS. **1A** and **1B**, in one or more embodiments, housing **104** of device **100** may include a channel **124**, which may extend from cavity **116** and terminate at tray **128** (shown in FIG. **1B**). In one or more embodiments, channel **124** may include one or more sections. For instance, and without limitation, channel **124** may have a first section **144** and a second section **148** (indicated by the dot-dash lines in FIG. **1B**), as discussed further below. As understood by one skilled in the art, channel **124** may include various shapes and sizes to accommodate different shaped and sized pills that may be stored in storage cavity **116** and traversed through channel **124**. For instance, and without limitation, channel **124** may be cylindrical (as shown in FIG. **2**). In one or more embodiments, channel **124** may be relatively straight. More specifically, second section **148** of channel **124** may be parallel and/or aligned with first section **144** of channel **124**. In other embodiments, channel **124** may be angled. For instance, and without limitation, channel **124** may include a right angle therein (as shown in FIG. **1B**) or an obtuse angle. More specifically, second section **148** of channel **124** may be oriented at an angle relative to first section **144** of channel **124**. In other embodiments, channel **124** may be curved. As previously mentioned, channel **124** may be defined by inner surface **140** of housing **104**.

(20) Still referring to FIGS. **1A** and **1B**, in one or more embodiments, housing **104** may include tray **128**, which may be configured to receive dispensed pill **120**. In one or more embodiments, tray **128** may be located near a bottom of housing **104**. As understood by one skilled in the art, tray **128** may include various shapes and sizes. For instance, and without limitation, tray **128** may include an elongated trough, as shown in FIGS. **1A** and **1B**. In other instances, and without limitation, tray **128** may include, for example, a dish, such as a semi-circular dish. In other instances, and without limitation, tray **128** may include a plurality of individualized trays so that each tray **128** may be designated to a corresponding cavity **116** and channel **124**. In one or more embodiments, tray **128** may be integrated into housing **104** or may be removable so that a user may, for example, choose a desired shape and size for tray **128**, may clean tray **128**, or may replace tray **128** if tray **128** is damaged.

(21) In one or more embodiments, tray **128** may include a security enclosure (not shown). For instance, and without limitation, to ensure an unintended individual does not take or tamper with dispensed pill **120**, tray **128** may include a sliding or hinged door. In one or more embodiments, security enclosure may be opened so that dispensed pill **120** may be retrieved from tray **128** by a pre-approved individual, such as the user or a caregiver, by using a security component, such as a fingerprint reader **152**, a passcode, or a physical key. For example, a user may place their thumb on fingerprint reader **152** to unlock a security enclosure of tray **128** and retrieve dispensed pill **120** from tray **128**. In one or more embodiments, tray **128** may be made of various materials. For example, tray **128** may include a polymer or a metal. In other embodiments, tray **128** may include a

flexible material, such as such as silicon, to prevent pill **120** from sliding or bouncing off tray **128** when it is ejected from channel **124** onto tray **128**.

(22) Still referring to FIGS. **1A** and **1B**, in one or more embodiments, device **100** may include dispensing element **108** (shown in FIG. **1B**), which is configured to dispense pill **120** from storage cavity **116** into tray **128** when actuated. In one or more embodiments, dispensing element may be at least partially disposed within recess **156** of housing **104**. Furthermore, dispensing element **108** may be disposed at least partially within channel **124**. For instance, and without limitation, dispensing element **108** in an exemplary embodiment may move into an engaged position so that channel **124** is divided into a first section **144** and a second section **148**, thus, blocking channel **124** and preventing any pills **120** from traversing through second section **148** of channel **124** to tray **128**. In a disengaged position, dispensing element **108** may be moved toward recess **156** so that channel **124** is opened and thus first section **144** and second section **148** of channel **124** are communicatively connected so that at least one pill **120** may traverse from first section **144** to second section **148**. In a disengaged position, dispensing element **108** may be moved so that channel **124** is partially open or completely opened, depending on the size and shape of pill **120**. Though dispensing element **108** is described as a retractable dispensing element, as understood by one skilled in the art, dispensing element may include various shapes, sizes, and mechanisms. For example, dispensing mechanism **108** may include a retractable plug, a spring-loaded plug, a rotatable body, a slotted piece, a rotatable wheel, or the like, as discussed further below.

(23) Still referring to FIGS. **1A** and **1B**, in one or more embodiments, device **100** may include a controller **160**. In one or more embodiments, controller **160** may be in communication with or control various components of device **100** or remote devices. For instance, and without limitation, controller **160** may be configured to actuate dispensing element **108** at a predetermined dispensing time. For example, controller **160** may be programmed to dispense a single pill at 7:15 PM EST. Thus, at 7:15 PM EST controller may actuate dispensing element **108** so that pill **120** may traverse through channel **124** and into tray **128** for retrieval by a user. Predetermined dispensing time may be recorded in memory of device **100** and/or accessible via electronic communication for device **100**; for instance, predetermined time may be set using interface elements of device **100** and/or a user device, and may be stored in user device and/or device **100**. Alternatively or additionally, device **100**, user device, and/or a remote device may generate predetermined dispensing time using a machine-learning algorithm and/or model as described below, which may be trained using training examples correlating past retrievals to retrieval times, user entries logging past retrievals and/or times, or the like.

(24) In one or more embodiments, controller **160** may include a user input and/or interface. For example, and without limitation, a user interface of controller **160** may include actuated components such as, for example, one or more buttons, knobs, joysticks, or slide bars. In another example, a user interface of controller **160** may include a display **136**, which may be a touchscreen that may be actuated to provide an input signal from a user. Display **136** may include a graphic user interface (GUI) that a user may navigate using prompts to select desired commands and thus provide certain input signals, as discussed further below.

(25) In one or more embodiments, controller **160** may include a memory component. In one or more embodiments, memory component may store information and data from various components of device **100**. A memory component may include various types of memory devices including, but not limited to, volatile and non-volatile memory devices. For example, a memory component may include a volatile memory device such as DRAM (Dynamic random-access memory), SRAM (static random-access memory), or the like. In another example, memory component may include non-volatile memory such as ROM (Read-only memory), flash memory, EEPROM (Electrically erasable read-only memory), or the like.

(26) In one or more embodiments, controller **160** may include a processor. A processor may be configured to interface and communicate with numerous components of device **100**. For instance,

without limitation, processor may communicate with a camera **164**, sensors, display **136**, status display **168**, LEDs, dispensing element **108**, storage cavities **116**, and the like to perform methods described in this disclosure. In other embodiments, a processor may execute software instructions provided by, for example, a memory.

(27) Still referring to FIGS. **1A** and **1B**, in one or more embodiments, device **100** may include one or more sensors, such as photoelectric sensor **172** and load cell sensor **176**. For instance, and without limitation, device **100** may use a sensor to detect when the quantity of the pill **120** in storage cavity **116** and/or channel **124** is below a predetermined quantity threshold. For example, photoelectric sensor **172** may be configured to detect when a number of pills in storage cavity **116** is less than a predetermined quantity threshold. Photoelectric sensor **172** may include, for example, a thru-beam sensor, a retroreflective sensor, a diffused sensor, or the like. In another example, load cell sensor **176** may be configured to detect when a number of pills in storage cavity **116** is less than a predetermined quantity threshold. Load cell sensor **176** may include, for example, a miniature load cell sensor, beam load cell sensor, a canister style load cell sensor, or the like. A pressure sensor (not shown) may also be used to determine the amount of pills in storage cavity **116** and/or channel **124**. Predetermined quantity threshold may depend, without limitation, on a type of pills, one or more elements of user data, and/or one or more elements of prescription information. Predetermined threshold may be retrieved from a database and/or lookup table using one or more elements of user data, and/or one or more elements of prescription information. Alternatively or additionally, a machine-learning process and/or model may be performed, generated, and/or loaded by device **100**, a user device, and/or a remote device; such machine-learning process and/or model may be trained using training data correlating one or more elements of user data, and/or one or more elements of prescription information to one or more predetermined threshold values, which training data may be received from one or more users, experts, or the like.

(28) Alternatively or additionally, and with continued reference to FIGS. **1A** and **1B**, predetermined threshold may be implemented using a membership function of a fuzzy set as described below, where a degree of membership above a threshold value and/or matching one or more inferential rules may indicate that the threshold is met and/or exceeded. Threshold value, coefficients, weights, and/or biases of inferential rules, and/or coefficients, weights and/or biases of membership function and/or functions may include user-entered and/or stored values, and/or may be determined using one or more machine-learning processes, based for instance on training data correlating one or more elements of user data, and/or one or more elements of prescription information to one or more coefficients, weights, and/or biases of inferential rules, and/or coefficients, weights and/or biases of membership function and/or functions, which training data may be received from one or more users, experts, or the like.

(29) In one or more embodiments, sensors **172,176** may be attached to housing **104**. For instance sensors **172,176** may be attached to inner surface **140** of housing **104**. For example, photoelectric sensor **172** may be attached to inner surface **140** and disposed within channel **124**. In another example, inner surface **140** may be a transparent material that allows photoelectric sensor **172** to be attached to the outside of inner surface **140** while still operating as intended. In one or more embodiments, sensors **172,176** may be positioned in various locations within cavity **116** and/or channel **124** to detect the amount of pills **120** within storage cavity **116** and/or channel **124**.

(30) In one or more exemplary embodiments, pill quantity datum may be detected by sensors **172,176** and transmitted to controller **160** as an output signal. Controller **160** may then determine whether the amount detected is less than a predetermined quantity threshold set by, for example, user or medical personnel. If controller **160** determines an amount of pills **120** is less than a predetermined amount, controller **160** may notify a user or remote personnel. Additionally, if pill **120** is a prescription medication, controller **160** may communicate with remote personnel, such as with a pharmacist, to request a refill of prescription medication.

(31) Though not shown, device **100** may include various other types of sensors. For instance, and

without limitation, device **100** may include a motion sensor, a temperature sensor, thermocouples, thermistors, thermometers, infrared sensors, resistance temperature sensors (RTDs), semiconductor based integrated circuits (IC), a combination thereof or another undisclosed sensor type, alone or in combination. For instance, and without limitation, a temperature sensor may be used to detect if the temperature within device **100** has risen above or fallen below predetermined temperature thresholds.

(32) Predetermined temperature threshold may depend, without limitation, on a type of pills, one or more elements of user data, and/or one or more elements of prescription information.

Predetermined temperature threshold may be retrieved from a database and/or lookup table using one or more elements of user data, and/or one or more elements of prescription information.

Alternatively or additionally, a machine-learning process and/or model may be performed, generated, and/or loaded by device **100**, a user device, and/or a remote device; such machine-learning process and/or model may be trained using training data correlating one or more elements of user data, and/or one or more elements of prescription information to one or more predetermined threshold values, which training data may be received from one or more users, experts, or the like.

(33) Alternatively or additionally, and with continued reference to FIGS. **1A** and **1B**, Predetermined temperature threshold may be implemented using a membership function of a fuzzy set as described below, where a degree of membership above a threshold value and/or matching one or more inferential rules may indicate that the threshold is met and/or exceeded. Threshold value, coefficients, weights, and/or biases of inferential rules, and/or coefficients, weights and/or biases of membership function and/or functions may include user-entered and/or stored values, and/or may be determined using one or more machine-learning processes, based for instance on training data correlating one or more elements of user data, and/or one or more elements of prescription information to one or more coefficients, weights, and/or biases of inferential rules, and/or coefficients, weights and/or biases of membership function and/or functions, which training data may be received from one or more users, experts, or the like.

(34) Still referring to FIGS. **1A** and **1B**, pills **120** may be sensitive to extreme temperatures and spoil or lose efficacy if exposed to such temperatures. In an exemplary embodiment, if temperature sensor detects that the temperature within cavity **116** is higher than a predetermined temperature threshold, then a user may be notified. Temperature, for the purposes of this disclosure, and as would be appreciated by someone of ordinary skill in the art, is a measure of the heat energy of a system. Heat energy is, at its core, the measure of kinetic energy of any or all matter present within a system. Temperature, as read by any number or combinations of sensors present on sense board **404**, may be measured in Fahrenheit (° F.), Celsius (° C.), Kelvin (° K), or another scale alone or in combination. The temperature measured by sensors may comprise electrical signals which are transmitted to appropriate destination wireless or through a wired connection. Outputs from sensors or any other component present within device **100** may be analog or digital. Controller **160** may convert output signals from sensors to a usable form. The usable form of output signals from sensors, through controller **160** and of a processor of controller **160** may be either digital, analog, a combination thereof or an otherwise unstated form. Processing may be configured to trim, offset, or otherwise compensate the outputs of the at least a sensor. Based on sensor output, the processor can determine the output to send to a downstream component. A processor can include signal amplification, operational amplifier (OpAmp), filter, digital/analog conversion, linearization circuit, current voltage change circuits, resistance change circuits, or the like.

(35) In one or more embodiments, device **100** may include a light-emitting diode (LED) indicator **180** to indicate a status of device **100**. In one or more embodiments, LED indicators **180** might indicate low medication. For instance, and without limitation, device **100** may include a plurality of LED indicators **180** configured to indicate that sensors, such as photoelectric sensor **172** and/or load cell sensor **176**, detect the number of pills **120** is less than the predetermined quantity threshold. Each storage cavity **116** may have a corresponding LED indicator **180** that shows when

that specific storage cavity **116** has a low pill quantity. Sensors **172,176** may be programmable and adjustable since a type of pills stored in each cavity **116** may vary. In one or more embodiments, a size and weight of an individual pill may be provided to controller **160** via, for example, display **136** or a remote electronic device so controller **160** may program sensors **172,176** and adjust parameters of sensors **172,176** accordingly. When sensors **172,176** detect that storage cavity **116** is low in pills **120**, then LED indicator **180** may illuminate to notify a user of the status of cavity **116**. For example, LED indicator **180** may illuminate to indicate that cavity **116** has a number of pills lower than the predetermined quantity threshold and a refill is required. In addition, status display **168** may show a notification regarding the pill quantity in cavity **116** and may also include a notification if any steps have been taken by device **100** to assist with the low number of pills, such as contacting a pharmacist with a refill request.

(36) In one or more embodiments, LED indicator **180** may include a dual-LED system. For example, as shown in FIG. **1A**, LED indicator **180** may have a first LED **184** and a second LED **188**. A dual-LED system may allow for a progressive status monitoring of a status of cavity **116** and/or channel **124**. For instance, and without limitation, first LED **184** may illuminate to indicate that storage cavity **116** is empty as detected by, for example, pressure sensor, and second LED **188** may illuminate to indicate that channel **124** is low or empty on pills **120** as detected by, for example, an eye sensor or photoelectric sensor **172**. First LED **184** and second LED **188** may be illuminated with the same color or varying colors. For example, first LED **184** may be yellow, and second LED may be red.

(37) Still referring to FIGS. **1A** and **1B**, in one or more embodiments, device **100** may include display **136**, which is configured show a user information and allows user to interface with device **100**. In one or more embodiments, display **136** may be attached to housing **104**. In one or more embodiments, display **136** may show information about any pill stored in any cavity **116** of device **100**. Information about a pill may include a name of a pill, a dosage of pill, an expiration date of pill, a next refill date of pill, a corresponding physician's name and contact information for pill, a refill alert for a pill, an image of pill, and so on.

(38) In one or more embodiments, display **136** may be used to contact a remote device or individual using a communication component **196** of device **100**, as discussed further below.

(39) In one or more embodiments, display **136** may indicate status of device **100**. For example, display **136** may show if device **100** is in a "secure" mode or an "unsecure" mode, as discussed further below. Display **136** may also show the status of cavity **116** and/or channel **124**. For example, display may show an expected refill date or if cavity **116** and/or channel **124** is low on pills or empty.

(40) In one or more embodiments, display **136** may provide menus and prompts that may be navigated to show various information or input various commands to be executed by controller **160**. For example, display **136** may show contact information of important personnel, such as a caregiver, primary care physician (PCP), or emergency personnel. In one or more exemplary embodiments, display **136** may be used during communication. For example, display **136** may show an image, such as a picture or video, of a remote individual during a telephonic call. For example, a physician may be contacted using communication component **196** and a live-stream video of the physician during the call may be shown on display **136**.

(41) In one or more embodiments, display **136** may be a touchscreen. In one or more embodiments, display **136** may be foldable or hinged so that screen may be angled according to a user's preference. Display **136** may be various types of electronic visual displays. For instance, and without limitation, display **136** may include a liquid crystal (LCD) display, a light-emitting diode (LED) display, a plasma display, a quantum dot display, or other various types of video or monitor displays generally known. As mentioned in this disclosure, a processor of controller **160** may be configured to display a captured image on display **136**.

(42) In one or more embodiments, device **100** may include a status display **168**. Status display **168**

may be attached to housing **104**. Status display **168** may show information regarding pills **120** in cavity **116**. Status display **168** may also show information of pill, such as name of the pill type, contents, or the like so that a user may, at any time, know what type of pills is in which cavity **116**. Information may include a name of a pill, a dosage of pill, an expiration date of pill, a next refill date of pill, a corresponding physician's name and contact information for pill, a refill alert for a pill, an image of pill, and so on. For instance, without limitation, status display **168** may provide a picture of dispensed pill **120** so that a user may know what pill to expect when retrieving pill **120** from tray **128**.

(43) Still referring to FIGS. **1A** and **1B**, device **100** may include camera **164**, which is configured to capture an image. In one or more embodiments, camera **164** may be configured to capture an image of a user. For instance, and without limitation, camera **164** may take a picture of a user when a motion sensor on a front of housing **104** and/or a pressure sensor on tray **128** detect that a dispensed pill **120** has been removed from tray **128**. The capturing of the image may be to ensure that a user is picking up dispensed pill **120** and not an unintended individual, such as a child. In another instance, and without limitations, when an alarm transducer alerts a user that pill **120** has been dispensed into tray **128**, camera may capture an image, such as a video recording, until dispensed pill **120** is retrieved. In other embodiments, when a user accesses tray **128** using, for example, fingerprint reader **152**, camera **164** may capture an image of user. In one or more embodiments, an image of user may, for example, be transmitted to a caregiver so caregiver may confirm that user picked up pill **120**, or the image may be stored in a memory of controller **160** for later retrieval by user or caregiver. In another exemplary embodiment, motion sensor **192** of device **100** may detect movement, the movement information signal may be sent to controller **160**, and controller **160** may command camera **164** to capture an image. As previously mentioned, as an additional security measure to ensure an unintended individual does not take or tamper with dispensed pill **120**, tray **128** may include one or more sliding or hinged doors that are locked until user placed finger on fingerprint reader **152** to gain access to tray **128**. Camera **164** may allow remote monitoring using camera **164**. For example, a caregiver may login and access camera **164** using a mobile device to monitor user.

(44) In one or more embodiments, camera **164** allows for capturing of an image. For example, camera **164** may be used to take a picture or a video image. A processor may be configured to process captured images from camera **164** and store the image data information in a memory of controller **160** or provide the image on display **136** or status display **168** for viewing by a user. For example, a user may capture an image of a pill using camera **164** so that a processor may store the image data in a memory and retrieve the image data to show the image on status display **168** when, for example, the type of pill captured in the image is dispensed by device **100**.

(45) In one or more embodiments, a motion sensor **192** may be positioned near camera **164** so that a picture is taken when a user is detected in front of device. For example, if motion sensor **192** detects a presence in front of device **100** while dispensed pill **120** is in tray **128**, camera **164** may take a picture as a security measure.

(46) In one or more embodiments, device **100** may include fingerprint reader **152**, which is configured to scan a fingerprint as a security measure. In one or more embodiments, a user may gain access to tray **128** using fingerprint reader **152**, as previously mentioned in this disclosure. Fingerprint reader **152** may also be used to gain access to an opening of housing **104** (not shown). For example, a caregiver may have their finger scanned to unlock panel **162** of a rear opening in housing **104**. In one or more embodiments, fingerprint reader may be attached to housing **104**. In one or more embodiments, fingerprint reader **152** may be integrated into housing **104**. In other embodiments, fingerprint reader **152** may be a separate component that is removeable from housing **104**. For instance, without limitation, fingerprint reader **152** may be removed from housing **104** so that fingerprint reader **152** may be exchanged for a new fingerprint reader with upgraded hardware and/or software. Thus, fingerprint reader **152** may be upgraded without replacing entire device **100**.

In other embodiments, fingerprint reader **152** may be removed to be inserted into an updated model of device **100**.

(47) Still referring to FIGS. **1A** and **1B**, device **100** may include speaker **166**, which is attached to housing **104**. In one or more embodiments, speaker **166** may be used to alert a user of when pill **120** has been dispensed into tray **128**, as previously described in this disclosure. Speaker **166** may use a preselected sound or may announce the time and/or a predetermined retrieval time before device **100** contacts, for example, a caregiver. In one or more embodiments, speaker **166** may also be used in conjunction with communication component **196** to communicate with others, as discussed in this disclosure. Alert may announce pill **120** is in tray **128**, if not picked up within, for example, 2 minutes of pill **120** being dispensed into tray **128** then device **100** may contact, for example, designated caregiver of user. A speaker **166** may project an alarm signal via sound and display **136** may project a visual alert. Speaker **166**, display **136**, LEDs **180-188**, and/or any other component of device **100** capable of outputting a signal for a user to detect may act as an alarm transducer. An “alarm transducer,” as used in this disclosure, is any component or set of components usable to convert an electronic or data signal into a user-detectable signal such as a display, audible noise, or the like. Alarm transducer may alternatively or additionally be any component of a user device which is suitable for use as an alarm transducer as described above.

(48) In one or more embodiments, controller **160** may include a communication component **196**. Communication component **196** may allow for device **100** to communicate the status of pill **120** to a remote device operated by remote personnel. For instance, and without limitation, communication component **196** may allow for communication between device **100** and a laptop, desktop, mobile phone, tablet, smart devices, or the like. Communication component **196** may use wired or wireless communication, as discussed in this disclosure. Communication component **196** may include various communication devices known in the art. Communication device may allow for interfacing between controller **160** and/or processor and various components of device **100** and/or communication may allow interfacing between controller **160** and/or processor and a remote device, such as a mobile phone or computer. Communication component **196** may also include a microphone. In an embodiment, communication component may include a computing device. Computing device may include any computing device as described in this disclosure, including without limitation a microcontroller, microprocessor, digital signal processor (DSP) and/or system on a chip (SoC) as described in this disclosure. Computing device may include, be included in, and/or communicate with a mobile device such as a mobile telephone or smartphone. Computing device may include a single computing device operating independently, or may include two or more computing device operating in concert, in parallel, sequentially or the like; two or more computing devices may be included together in a single computing device or in two or more computing devices. Computing device may interface or communicate with one or more additional devices as described below in further detail via a network interface device. Network interface device may be utilized for connecting computing device to one or more of a variety of networks, and one or more devices. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. A network may employ a wired and/or a wireless mode of communication. In general, any network topology may be used. Information (e.g., data, software and the like) may be communicated to and/or from a computer and/or a computing device. Computing device may include but is not limited to, for example, a computing device or cluster of computing devices in a first location and a second computing device or cluster of computing devices in a second location.

Computing device may include one or more computing devices dedicated to data storage, security, distribution of traffic for load balancing, and the like. Computing device may distribute one or more computing tasks as described below across a plurality of computing devices of computing device, which may operate in parallel, in series, redundantly, or in any other manner used for distribution of tasks or memory between computing devices. Computing device may be implemented using a “shared nothing” architecture in which data is cached at the worker, in an embodiment, this may enable scalability of system **100** and/or computing device.

(49) With continued reference to FIG. **1**, computing device may be designed and/or configured to perform any method, method step, or sequence of method steps in any embodiment described in this disclosure, in any order and with any degree of repetition. For instance, computing device may be configured to perform a single step or sequence repeatedly until a desired or commanded outcome is achieved; repetition of a step or a sequence of steps may be performed iteratively and/or recursively using outputs of previous repetitions as inputs to subsequent repetitions, aggregating inputs and/or outputs of repetitions to produce an aggregate result, reduction or decrement of one or more variables such as global variables, and/or division of a larger processing task into a set of iteratively addressed smaller processing tasks. Computing device may perform any step or sequence of steps as described in this disclosure in parallel, such as simultaneously and/or substantially simultaneously performing a step two or more times using two or more parallel threads, processor cores, or the like; division of tasks between parallel threads and/or processes may be performed according to any protocol suitable for division of tasks between iterations. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various ways in which steps, sequences of steps, processing tasks, and/or data may be subdivided, shared, or otherwise dealt with using iteration, recursion, and/or parallel processing.

(50) In one or more embodiments, communication component **196** may be configured to facilitate a notifying of remote personnel if dispensed pill **120** has not been retrieved from tray **128** within a predetermined retrieval time. In one or more embodiments, a user may use communication component **196** to contact a caregiver, a pharmacist, a physician, emergency personnel, and so on. In other embodiments, a user may use communication component **196** to contact any other contact, such as a family member or friend. Predetermined retrieval time may depend, without limitation, on a type of pills, one or more elements of user data, and/or one or more elements of prescription information. Predetermined retrieval time may be retrieved from a database and/or lookup table using one or more elements of user data, and/or one or more elements of prescription information. Alternatively or additionally, a machine-learning process and/or model may be performed, generated, and/or loaded by device **100**, a user device, and/or a remote device; such machine-learning process and/or model may be trained using training data correlating one or more elements of user data, and/or one or more elements of prescription information to one or more predetermined threshold values, which training data may be received from one or more users, experts, or the like.

(51) In one or more embodiments, communication component **196** may be used for automated communication. For instance, and without limitation, controller **160** may use communication component **196** to contact a pharmacist after sensors **172,176** detect that a storage cavity **116** is low on a prescription medication and that a refill is required. A pharmacist may then receive the notification and fulfill the request for the new refill. Similarly, a pharmacist may then contact communication component **196** using a remote device and/or interface to notify a user that the request has been completed and that the refill is ready for pickup or being prepared for delivery.

(52) In the case of a new prescription being required, communication component **196** may be used to contact the physician of the user and/or leave a notification for the user on status display **168** regarding a new prescription requirement.

(53) In another exemplary embodiment, controller **160** may use communication component **196** to notify, for example, a caregiver, if a retrieval sensor, such as a pressure sensor, of tray **128**, detects that dispensed pill **120** has not been retrieved by a user in the predetermined retrieval time. A

retrieval sensor may be a photoelectric sensor, load cell sensor, or the like.

(54) In one or more embodiments, communication component **196** may be used to provide an alarm, for instance via an alarm transducer, to a user, physician, and/or caregiver's electronic device to alert the user that one or more pills has been dispensed into tray **128**. The predetermined retrieval time may also be displayed on the user's electronic device with an alarm signal. For example, a caregiver of the user may receive a notification on their mobile device via a graphic user interface that indicates pill **120** has been dispensed into tray **128**. The information related to the pill may also be provided in the alert. Subsequently, a notification may be updated by telling the caregiver that the user has retrieved dispensed pill **120** from tray **128**.

(55) Still referring to FIGS. **1A** and **1B**, device **100** may include a security component. In one or more embodiments, a security component may include a mechanical or electrical lock. For instance, and without limitation, a security component may include a physical lock **170** that requires a key to be opened. For example, storage cavity **116** may be locked with a physical lock to prevent undesired access to pills **120** contained within cavity **116**. In other embodiments, display **136** may require a passcode be entered in order for access to cavity **116** and/or tray **128** to be gained. In one or more embodiments, a security component may include a security sensor (not shown) configured to detect when storage cavity **116** has been accessed or a physical lock has been tampered with. In one or more embodiments, a security component may include fingerprint reader **152**, which may be used to unlock device **100**, as previously described in this disclosure. For example, display **136** may prompt a user to place their thumb on fingerprint reader **152** to unlock panel **162** of device **100** and gain access to storage cavities **116** and/or tray **128**.

(56) In one or more embodiments, device **100** may be programmed to have a “secured” or “unsecured” mode. For example, if device is in a secured mode, cavities **116** cannot be accessed with using, for example, fingerprint reader **152** or unlocking lock **170**, and communication component **196** may send out an alert if tampering with device **100** is detected by a security component. If device **100** is in an unsecured mode, then storage cavities **116** and/or tray **128** may be readily accessed without using a security component.

(57) In one or more embodiments, a security component may include a “fail-safe” mode. In a fail-safe mode, if device **100** loses power while in a secured mode, device **100** will remain secure to prevent tampering with storage cavities **116**.

(58) In one or more embodiments, device may be powered by a battery or by using a plug. In one or more embodiments, a rechargeable lithium-ion battery may be used as a power source. In one or more embodiments, device **100** may have a plug and be compatible with 120 V electrical systems. In one or more embodiments, battery may be used as a backup power source in case power to a plug of device **100** is no longer supplied, such as during a power outage. In case of complete power failure of device **100**, a key may be used to unlock housing panel **162** and access pills **120**.

(59) Now referring to FIG. **2**, a partially transparent isometric view of an exemplary embodiment of device **100** with a plurality of storage cavities **116** is shown in accordance with one or more embodiments of the present disclosure. In one or more embodiments, device **100** may include a plurality of storage cavities **116**, where each cavity **116** may be filled with a different type of pill **120** that may be dispensed at various predetermined times. In one or more embodiments, each storage cavity **116** may be communicatively connected to a corresponding status display **168**, LED indicator **180**, and sensor, such as sensors **172,176**.

(60) FIGS. **3A-3D** illustrate various exemplary embodiments of dispensing element **108** in accordance with one or more embodiments of the present disclosure. FIGS. **3A** and **3B** are partial cross-sectional views of exemplary embodiments of dispensing element **108** as a retractable dispensing element. As shown in FIG. **3A**, dispensing element **108** may include a plug **308** moveably attached to an arm **304**. Arm **304** may be attached to a lumen **312** by a pivot point **316**. Rotation of pivot pint **316** may move arm **304** and, thus, move plug **308** in a relatively lateral movement, such as toward and away from lumen **312**. For example, rotation of pivot point **316** by,

for example, controller **160** moves plug **308** fore and aft within recess **156** (shown in FIG. **1B**). As plug **308** moves fore and aft within recess **156**, pills **120** may be dispensed. For instance, and without limitation, when pivot point **316** is rotated in a first direction, as indicated by arrow **320**, plug **308** may be moved toward lumen **312**, as indicated by arrow **324**, creating a distance d within channel **124** (between inner surface **140** and plug **308**) wide enough to allow a single pill to pass plug **308** and traverse through channel **124** to tray **128**.

(61) As shown in FIG. **3B**, dispensing element **108** may be a spring-loaded retractable element. Plug **328** may be connected to an arm **332**, which is fixedly connected to a platform **336** that is slidably disposed within a lumen **340**. Platform **336** may be abutting a spring **388** that may be compressed or expanded. Plug **328** may be moved fore and aft within recess **156** (shown in FIG. **1B**) when platform **336** is moved fore and aft within lumen **340**. When plug **328** is retracted toward lumen **340** (as indicated by arrow **348**), a distance d' is created between inner surface **140** and plug **328** that is wide enough for a single pill to pass plug **328** and traverse through channel **124** and into tray **128**.

(62) As shown in FIGS. **3C** and **3D**, dispensing element **108** may also be a rotational dispensing element. As shown in FIG. **3C**, dispensing element **108** may include a body **352**, which has a thread **356** extending therefrom. Thread **356** may have a root **360** between crests **364** of thread **356**. A width w of root **360** may be such that pill **120** may be seated within root **360** and abutting flanks **368** of thread **356**. Thread **356** may also include a height h that allows pill **120** to be recessed within thread **356** and thus below height h of thread **356**. Body **352** may be rotated about a central axis A of body **352** to move pill **120** through channel **124** and ultimately into tray **128**. For instance, and without limitation, body **352** may be rotated about central axis A so that pills **120** disposed within thread **356** are advanced in a direction parallel to central axis A with each revolution of body **352** about central axis A . Thus, in an exemplary embodiment, a single pill may be traversed through channel **124** with, for example, each quarter revolution of body **352**.

(63) As shown in FIG. **3D**, dispensing element **108** may be a rotatable dispensing element with a body **376**, which includes an indentation **380**. Indentation **380** may be sized so that pill **120** may be at least partially seated within depth D of indentation **380** so that when body **376** rotates about a central axis B of body **376**, pill **120** may be rotated about central axis B . Thus, in an exemplary embodiment, when body **376** is rotated, for example, a half rotation, pill **120** may be moved from a first section **144** of channel **124** to a second section **148** of channel **124** and into tray **128**. The rotatable dispensing element **108** of FIG. **3D** may also be combined with a retractable component of, for example, dispensing elements **108** of FIGS. **3A** and **3B**. For example, pill **120** may be completely seated into indentation **380**, body **376** may then be moved away from recess **156** and rotated so that pill is then dropped into second section **148** of channel **124** (shown in FIG. **1B**).

(64) As understood by one skilled, dispensing element **108** may be various sizes and shapes without changing the scope or spirit of the invention. For example, dispensing element **108** may be a pin wheel, two slidable surfaces with slots that may be aligned to allow a pill to traverse through the aligned slots of the surfaces, or the like. Also, understood by one skilled in the art, the proportions of dispensing element **108** may be adjusted according to the size and shape of pill **120**. In one or more embodiments, dispensing element **108** may be interchangeable and thus removed and replaced with other types of dispensing elements as desired.

(65) FIG. **4** is a flow chart showing a method **400** of automatically dispensing pill **120** using programmable pill dispensing device **100**. As shown in block **405**, method **400** includes storing pills **120** in storage cavity **116**, which is defined by housing **104** of device **100**.

(66) As shown in block **410**, method **400** includes actuating dispensing element **108** of device **100** at a predetermined time using controller **160**.

(67) As shown in block **415**, method includes dispensing, upon actuation of dispensing element **108**, pill **120** through channel **124**, which extends from storage cavity **116** and is connected to tray **128**, so that pill **120** traverses through channel **124** and into tray **128**. Method **400** may further

include detecting, by at least sensor, when a number of pills **120** in storage cavity **116** is less than a predetermined quantity threshold. In one or more embodiments, sensor may include at least sensors **172,176**. Method **400** may further include indicating, by light-emitting diode (LED) indicator **180**, that a sensor detects a number of pills **120** is less than the predetermined quantity threshold.

(68) As shown in block **420**, method **400** includes alerting, user an alarm transducer, a user that pill **120** is dispensed and ready for retrieval by a user. Method **400** may further include notifying, by communication component **196**, a remote personnel that dispensed pill **120** has not been retrieved from tray **128** within a redetermined retrieval time. Method **400** may further include, by a communication component, facilitating communication between device **100** and remote personnel.

(69) Examples of a computing device include, but are not limited to, an electronic book reading device, a computer workstation, a terminal computer, a server computer, a handheld device (e.g., a tablet computer, a smartphone, etc.), a web appliance, a network router, a network switch, a network bridge, any machine capable of executing a sequence of instructions that specify an action to be taken by that machine, and any combinations thereof. In one example, a computing device may include and/or be included in a kiosk.

(70) Referring now to FIG. **5**, an exemplary embodiment of a machine-learning module **500** that may perform one or more machine-learning processes as described in this disclosure is illustrated. Machine-learning module may perform determinations, classification, and/or analysis steps, methods, processes, or the like as described in this disclosure using machine learning processes. A “machine learning process”, as used in this disclosure, is a process that automatedly uses training data **504** to generate an algorithm that will be performed by a computing device/module to produce outputs **508** given data provided as inputs **512**; this is in contrast to a non-machine learning software program where the commands to be executed are determined in advance by a user and written in a programming language.

(71) Still referring to FIG. **5**, “training data,” as used herein, is data containing correlations that a machine-learning process may use to model relationships between two or more categories of data elements. For instance, and without limitation, training data **504** may include a plurality of data entries, each entry representing a set of data elements that were recorded, received, and/or generated together; data elements may be correlated by shared existence in a given data entry, by proximity in a given data entry, or the like. Multiple data entries in training data **504** may evince one or more trends in correlations between categories of data elements; for instance, and without limitation, a higher value of a first data element belonging to a first category of data element may tend to correlate to a higher value of a second data element belonging to a second category of data element, indicating a possible proportional or other mathematical relationship linking values belonging to the two categories. Multiple categories of data elements may be related in training data **504** according to various correlations; correlations may indicate causative and/or predictive links between categories of data elements, which may be modeled as relationships such as mathematical relationships by machine-learning processes as described in further detail below. Training data **504** may be formatted and/or organized by categories of data elements, for instance by associating data elements with one or more descriptors corresponding to categories of data elements. As a non-limiting example, training data **504** may include data entered in standardized forms by persons or processes, such that entry of a given data element in a given field in a form may be mapped to one or more descriptors of categories. Elements in training data **504** may be linked to descriptors of categories by tags, tokens, or other data elements; for instance, and without limitation, training data **604** may be provided in fixed-length formats, formats linking positions of data to categories such as comma-separated value (CSV) formats and/or self-describing formats such as extensible markup language (XML), JavaScript Object Notation (JSON), or the like, enabling processes or devices to detect categories of data.

(72) Alternatively or additionally, and continuing to refer to FIG. **5**, training data **504** may include one or more elements that are not categorized; that is, training data **504** may not be formatted or

contain descriptors for some elements of data. Machine-learning algorithms and/or other processes may sort training data **504** according to one or more categorizations using, for instance, natural language processing algorithms, tokenization, detection of correlated values in raw data and the like; categories may be generated using correlation and/or other processing algorithms. As a non-limiting example, in a corpus of text, phrases making up a number “n” of compound words, such as nouns modified by other nouns, may be identified according to a statistically significant prevalence of n-grams containing such words in a particular order; such an n-gram may be categorized as an element of language such as a “word” to be tracked similarly to single words, generating a new category as a result of statistical analysis. Similarly, in a data entry including some textual data, a person's name may be identified by reference to a list, dictionary, or other compendium of terms, permitting ad-hoc categorization by machine-learning algorithms, and/or automated association of data in the data entry with descriptors or into a given format. The ability to categorize data entries automatically may enable the same training data **504** to be made applicable for two or more distinct machine-learning algorithms as described in further detail below. Training data **504** used by machine-learning module **500** may correlate any input data as described in this disclosure to any output data as described in this disclosure.

(73) Further referring to FIG. 5, training data may be filtered, sorted, and/or selected using one or more supervised and/or unsupervised machine-learning processes and/or models as described in further detail below; such models may include without limitation a training data classifier **516**. Training data classifier **516** may include a “classifier,” which as used in this disclosure is a machine-learning model as defined below, such as a mathematical model, neural net, or program generated by a machine learning algorithm known as a “classification algorithm,” as described in further detail below, that sorts inputs into categories or bins of data, outputting the categories or bins of data and/or labels associated therewith. A classifier may be configured to output at least a datum that labels or otherwise identifies a set of data that are clustered together, found to be close under a distance metric as described below, or the like. Machine-learning module **500** may generate a classifier using a classification algorithm, defined as a process whereby a computing device and/or any module and/or component operating thereon derives a classifier from training data **504**. Classification may be performed using, without limitation, linear classifiers such as without limitation logistic regression and/or naive Bayes classifiers, nearest neighbor classifiers such as k-nearest neighbors' classifiers, support vector machines, least squares support vector machines, fisher's linear discriminant, quadratic classifiers, decision trees, boosted trees, random forest classifiers, learning vector quantization, and/or neural network-based classifiers.

(74) Still referring to FIG. 5, machine-learning module **500** may be configured to perform a lazy-learning process **520** and/or protocol, which may alternatively be referred to as a “lazy loading” or “call-when-needed” process and/or protocol, may be a process whereby machine learning is conducted upon receipt of an input to be converted to an output, by combining the input and training set to derive the algorithm to be used to produce the output on demand. For instance, an initial set of simulations may be performed to cover an initial heuristic and/or “first guess” at an output and/or relationship. As a non-limiting example, an initial heuristic may include a ranking of associations between inputs and elements of training data **504**. Heuristic may include selecting some number of highest-ranking associations and/or training data **504** elements. Lazy learning may implement any suitable lazy learning algorithm, including without limitation a K-nearest neighbors' algorithm, a lazy naïve Bayes algorithm, or the like; persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various lazy-learning algorithms that may be applied to generate outputs as described in this disclosure, including without limitation lazy learning applications of machine-learning algorithms as described in further detail below.

(75) Alternatively or additionally, and with continued reference to FIG. 5, machine-learning processes as described in this disclosure may be used to generate machine-learning models **524**. A “machine-learning model,” as used in this disclosure, is a mathematical and/or algorithmic

representation of a relationship between inputs and outputs, as generated using any machine-learning process including without limitation any process as described above and stored in memory; an input is submitted to a machine-learning model **524** once created, which generates an output based on the relationship that was derived. For instance, and without limitation, a linear regression model, generated using a linear regression algorithm, may compute a linear combination of input data using coefficients derived during machine-learning processes to calculate an output datum. As a further non-limiting example, a machine-learning model **524** may be generated by creating an artificial neural network, such as a convolutional neural network comprising an input layer of nodes, one or more intermediate layers, and an output layer of nodes. Connections between nodes may be created via the process of “training” the network, in which elements from a training data **504** set are applied to the input nodes, a suitable training algorithm (such as Levenberg-Marquardt, conjugate gradient, simulated annealing, or other algorithms) is then used to adjust the connections and weights between nodes in adjacent layers of the neural network to produce the desired values at the output nodes. This process is sometimes referred to as deep learning.

(76) Still referring to FIG. 5, machine-learning algorithms may include at least a supervised machine-learning process **528**. At least a supervised machine-learning process **528**, as defined herein, include algorithms that receive a training set relating a number of inputs to a number of outputs, and seek to find one or more mathematical relations relating inputs to outputs, where each of the one or more mathematical relations is optimal according to some criterion specified to the algorithm using some scoring function. For instance, a supervised learning algorithm may include user signals as described above as inputs, autonomous functions as outputs, and a scoring function representing a desired form of relationship to be detected between inputs and outputs; scoring function may, for instance, seek to maximize the probability that a given input and/or combination of elements inputs is associated with a given output to minimize the probability that a given input is not associated with a given output. Scoring function may be expressed as a risk function representing an “expected loss” of an algorithm relating inputs to outputs, where loss is computed as an error function representing a degree to which a prediction generated by the relation is incorrect when compared to a given input-output pair provided in training data **504**. Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various possible variations of at least a supervised machine-learning process **528** that may be used to determine relation between inputs and outputs. Supervised machine-learning processes may include classification algorithms as defined above.

(77) Further referring to FIG. 5, machine learning processes may include at least an unsupervised machine-learning processes **532**. An unsupervised machine-learning process, as used herein, is a process that derives inferences in datasets without regard to labels; as a result, an unsupervised machine-learning process may be free to discover any structure, relationship, and/or correlation provided in the data. Unsupervised processes may not require a response variable; unsupervised processes may be used to find interesting patterns and/or inferences between variables, to determine a degree of correlation between two or more variables, or the like.

(78) Still referring to FIG. 5, machine-learning module **500** may be designed and configured to create a machine-learning model **524** using techniques for development of linear regression models. Linear regression models may include ordinary least squares regression, which aims to minimize the square of the difference between predicted outcomes and actual outcomes according to an appropriate norm for measuring such a difference (e.g., a vector-space distance norm); coefficients of the resulting linear equation may be modified to improve minimization. Linear regression models may include ridge regression methods, where the function to be minimized includes the least-squares function plus term multiplying the square of each coefficient by a scalar amount to penalize large coefficients. Linear regression models may include least absolute shrinkage and selection operator (LASSO) models, in which ridge regression is combined with multiplying the least-squares term by a factor of 1 divided by double the number of samples. Linear

regression models may include a multi-task lasso model wherein the norm applied in the least-squares term of the lasso model is the Frobenius norm amounting to the square root of the sum of squares of all terms. Linear regression models may include the elastic net model, a multi-task elastic net model, a least angle regression model, a LARS lasso model, an orthogonal matching pursuit model, a Bayesian regression model, a logistic regression model, a stochastic gradient descent model, a perceptron model, a passive aggressive algorithm, a robustness regression model, a Huber regression model, or any other suitable model that may occur to persons skilled in the art upon reviewing the entirety of this disclosure. Linear regression models may be generalized in an embodiment to polynomial regression models, whereby a polynomial equation (e.g., a quadratic, cubic or higher-order equation) providing a best predicted output/actual output fit is sought; similar methods to those described above may be applied to minimize error functions, as will be apparent to persons skilled in the art upon reviewing the entirety of this disclosure.

(79) Continuing to refer to FIG. 5, machine-learning algorithms may include, without limitation, linear discriminant analysis. Machine-learning algorithm may include quadratic discriminate analysis. Machine-learning algorithms may include kernel ridge regression. Machine-learning algorithms may include support vector machines, including without limitation support vector classification-based regression processes. Machine-learning algorithms may include stochastic gradient descent algorithms, including classification and regression algorithms based on stochastic gradient descent. Machine-learning algorithms may include nearest neighbors' algorithms. Machine-learning algorithms may include Gaussian processes such as Gaussian Process Regression. Machine-learning algorithms may include cross-decomposition algorithms, including partial least squares and/or canonical correlation analysis. Machine-learning algorithms may include naïve Bayes methods. Machine-learning algorithms may include algorithms based on decision trees, such as decision tree classification or regression algorithms. Machine-learning algorithms may include ensemble methods such as bagging meta-estimator, forest of randomized trees, AdaBoost, gradient tree boosting, and/or voting classifier methods. Machine-learning algorithms may include neural net algorithms, including convolutional neural net processes.

(80) In one or more exemplary embodiments, machine learning may be used to, for example, detect when a user is noncompliant or when a user misses a dose. For example, and without limitation, an input **512**, such as type of pill stored in cavity **116**, may be inputted into machine-learning module **500**, which may in turn provide output **508**, such as predetermined times that a pill should be administer and thus dispensed from device **100**, predetermined retrieval times, and/or alarm settings. In one or more exemplary embodiments, training data set **504** may include data associated with the type of pill, such data regarding a dosage amount, a visual representation of the pill, chemical composition of the pill, and so on. In one or more embodiments, machine-learning module **500** may also be used to detect when a user takes a dose, or accesses pill **120** of storage cavity **116**, outside of the predetermined time. In other embodiments, machine-learning module **500** may be used to detect when pill **120** has not been retrieved from device **100** and remote personnel must be notified and/or camera **164** should be activated.

(81) Referring now to FIG. 6, an exemplary embodiment of fuzzy set comparison **600** is illustrated. A first fuzzy set **604** may be represented, without limitation, according to a first membership function **608** representing a probability that an input falling on a first range of values **616** is a member of the first fuzzy set **604**, where the first membership function **608** has values on a range of probabilities such as without limitation the interval $[0,1]$, and an area beneath the first membership function **608** may represent a set of values within first fuzzy set **604**. Although first range of values **612** is illustrated for clarity in this exemplary depiction as a range on a single number line or axis, first range of values **612** may be defined on two or more dimensions, representing, for instance, a Cartesian product between a plurality of ranges, curves, axes, spaces, dimensions, or the like. First membership function **608** may include any suitable function mapping first range **612** to a probability interval, including without limitation a triangular function defined

by two linear elements such as line segments or planes that intersect at or below the top of the probability interval. As a non-limiting example, triangular membership function may be defined as:

$$0, \text{ for } x > c \text{ and } x < a$$

$$(82) y(x, a, b, c) = \begin{cases} \frac{x-a}{b-a}, & \text{for } a \leq x < b \\ \frac{c-x}{c-b}, & \text{if } b < x \leq c \end{cases}$$

a trapezoidal membership function may be defined as:

$$(83) y(x, a, b, c, d) = \max(\min(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}), 0)$$

a sigmoidal function may be defined as:

$$(84) y(x, a, c) = \frac{1}{1 + e^{-a(x-c)}}$$

a Gaussian membership function may be defined as:

$$(85) y(x, c,) = e^{-\frac{1}{2}(\frac{x-c}{\sigma})^2}$$

and a bell membership function may be defined as:

$$(86) y(x, a, b, c,) = [1 + \text{Math.} \frac{x-c}{a-b} \cdot \text{Math.} \frac{2b}{a-b}]^{-1}$$

Persons skilled in the art, upon reviewing the entirety of this disclosure, will be aware of various alternative or additional membership functions that may be used consistently with this disclosure. A fuzzy set may include Cartesian products of two or more fuzzy sets as described above.

(87) First fuzzy set **604** may represent any value or combination of values as described above, A second fuzzy set **616**, which may represent any value which may be represented by first fuzzy set **604**, may be defined by a second membership function **620** on a second range **624**; second range **624** may be identical and/or overlap with first range **612** and/or may be combined with first range via Cartesian product or the like to generate a mapping permitting evaluation overlap of first fuzzy set **604** and second fuzzy set **616**. Where first fuzzy set **604** and second fuzzy set **616** have a region **628** that overlaps, first membership function **608** and second membership function **620** may intersect at a point **632** representing a probability, as defined on probability interval, of a match between first fuzzy set **604** and second fuzzy set **616**. Alternatively or additionally, a single value of first and/or second fuzzy set may be located at a locus **636** on first range **612** and/or second range **624**, where a probability of membership may be taken by evaluation of first membership function **608** and/or second membership function **620** at that range point. A probability at **628** and/or **632** may be compared to a threshold **640** to determine whether a positive match is indicated. Threshold **640** may, in a non-limiting example, represent a degree of match between first fuzzy set **604** and second fuzzy set **616**, and/or single values therein with each other or with either set, which is sufficient for purposes of the matching process; for instance, threshold may indicate a sufficient degree of overlap between fields and/or parameters represented by one or more fuzzy sets and/or Cartesian products thereof for combination to occur as described above. There may be multiple thresholds. Each threshold may be established by one or more user inputs. Alternatively or additionally, each threshold may be tuned by a machine-learning and/or statistical process, for instance and without limitation as described above.

(88) It is to be noted that any one or more of the aspects and embodiments described herein may be implemented using one or more machines. For example, machines such as computing devices that are utilized as a user computing device for an electronic document, one or more server devices, such as a document server, or the like, may be implemented and programmed according to the teachings of this disclosure. Appropriate software coding can readily be prepared by skilled programmers based on the teachings of the present disclosure, as will be apparent to those of ordinary skill in the software art. Aspects and implementations discussed above employing software and/or software modules may also include appropriate hardware for assisting in the implementation of the machine executable instructions of the software and/or software module.

(89) Such software may be a computer program product that employs a machine-readable storage medium. A machine-readable storage medium may be any medium that is capable of storing and/or

encoding a sequence of instructions for execution by a machine (e.g., a computing device) and that causes the machine to perform any one of the methodologies and/or embodiments described herein. Examples of a machine-readable storage medium include, but are not limited to, a magnetic disk, an optical disc (e.g., CD, CD-R, DVD, DVD-R, etc.), a magneto-optical disk, a read-only memory “ROM” device, a random-access memory “RAM” device, a magnetic card, an optical card, a solid-state memory device, an EPROM, an EEPROM, and any combinations thereof. A machine-readable medium, as used herein, is intended to include a single medium as well as a collection of physically separate media, such as, for example, a collection of compact discs or one or more hard disk drives in combination with a computer memory. As used herein, a machine-readable storage medium does not include transitory forms of signal transmission.

(90) Such software may also include information (e.g., data) carried as a data signal on a data carrier, such as a carrier wave. For example, machine-executable information may be included as a data-carrying signal embodied in a data carrier in which the signal encodes a sequence of instruction, or portion thereof, for execution by a machine (e.g., a computing device) and any related information (e.g., data structures and data) that causes the machine to perform any one of the methodologies and/or embodiments described herein.

(91) FIG. 7 shows a diagrammatic representation of one embodiment of a computing device in the exemplary form of a computer system **700** within which a set of instructions for causing a control system to perform any one or more of the aspects and/or methodologies of the present disclosure may be executed. It is also contemplated that multiple computing devices may be utilized to implement a specially configured set of instructions for causing one or more of the devices to perform any one or more of the aspects and/or methodologies of the present disclosure. Computer system **700** includes a processor **704** and a memory **708** that communicate with each other, and with other components, via a bus **712**. Bus **712** may include any of several types of bus structures including, but not limited to, a memory bus, a memory controller, a peripheral bus, a local bus, and any combinations thereof, using any of a variety of bus architectures.

(92) Memory **708** may include various components (e.g., machine-readable media) including, but not limited to, a random-access memory component, a read only component, and any combinations thereof. In one example, a basic input/output system **716** (BIOS), including basic routines that help to transfer information between elements within computer system **700**, such as during start-up, may be stored in memory **708**. Memory **708** may also include (e.g., stored on one or more machine-readable media) instructions (e.g., software) **720** embodying any one or more of the aspects and/or methodologies of the present disclosure. In another example, memory **708** may further include any number of program modules including, but not limited to, an operating system, one or more application programs, other program modules, program data, and any combinations thereof.

(93) Computer system **700** may also include a storage device **724**. Examples of a storage device (e.g., storage device **724**) include, but are not limited to, a hard disk drive, a magnetic disk drive, an optical disc drive in combination with an optical medium, a solid-state memory device, and any combinations thereof. Storage device **724** may be connected to bus **712** by an appropriate interface (not shown). Example interfaces include, but are not limited to, SCSI, advanced technology attachment (ATA), serial ATA, universal serial bus (USB), IEEE 1394 (FIREWIRE), and any combinations thereof. In one example, storage device **724** (or one or more components thereof) may be removably interfaced with computer system **700** (e.g., via an external port connector (not shown)). Particularly, storage device **724** and an associated machine-readable medium **728** may provide nonvolatile and/or volatile storage of machine-readable instructions, data structures, program modules, and/or other data for computer system **700**. In one example, software **720** may reside, completely or partially, within machine-readable medium **728**. In another example, software **720** may reside, completely or partially, within processor **704**.

(94) Computer system **700** may also include an input device **732**. In one example, a user of computer system **700** may enter commands and/or other information into computer system **700** via

input device **732**. Examples of an input device **732** include, but are not limited to, an alpha-numeric input device (e.g., a keyboard), a pointing device, a joystick, a gamepad, an audio input device (e.g., a microphone, a voice response system, etc.), a cursor control device (e.g., a mouse), a touchpad, an optical scanner, a video capture device (e.g., a still camera, a video camera), a touchscreen, and any combinations thereof. Input device **732** may be interfaced to bus **712** via any of a variety of interfaces (not shown) including, but not limited to, a serial interface, a parallel interface, a game port, a USB interface, a FIREWIRE interface, a direct interface to bus **712**, and any combinations thereof. Input device **732** may include a touch screen interface that may be a part of or separate from display **736**, discussed further below. Input device **732** may be utilized as a user selection device for selecting one or more graphical representations in a graphical interface as described above.

(95) A user may also input commands and/or other information to computer system **700** via storage device **724** (e.g., a removable disk drive, a flash drive, etc.) and/or network interface device **740**. A network interface device, such as network interface device **740**, may be utilized for connecting computer system **700** to one or more of a variety of networks, such as network **744**, and one or more remote devices **748** connected thereto. Examples of a network interface device include, but are not limited to, a network interface card (e.g., a mobile network interface card, a LAN card), a modem, and any combination thereof. Examples of a network include, but are not limited to, a wide area network (e.g., the Internet, an enterprise network), a local area network (e.g., a network associated with an office, a building, a campus or other relatively small geographic space), a telephone network, a data network associated with a telephone/voice provider (e.g., a mobile communications provider data and/or voice network), a direct connection between two computing devices, and any combinations thereof. A network, such as network **744**, may employ a wired and/or a wireless mode of communication. In general, any network topology may be used. Information (e.g., data, software **720**, etc.) may be communicated to and/or from computer system **700** via network interface device **740**.

(96) Computer system **700** may further include a video display adapter **752** for communicating a displayable image to a display device, such as display device **736**. Examples of a display device include, but are not limited to, a liquid crystal display (LCD), a cathode ray tube (CRT), a plasma display, a light emitting diode (LED) display, and any combinations thereof. Display adapter **752** and display device **736** may be utilized in combination with processor **704** to provide graphical representations of aspects of the present disclosure. In addition to a display device, computer system **700** may include one or more other peripheral output devices including, but not limited to, an audio speaker, a printer, and any combinations thereof. Such peripheral output devices may be connected to bus **712** via a peripheral interface **756**. Examples of a peripheral interface include, but are not limited to, a serial port, a USB connection, a FIREWIRE connection, a parallel connection, and any combinations thereof.

(97) The foregoing has been a detailed description of illustrative embodiments of the invention. Various modifications and additions can be made without departing from the spirit and scope of this invention. Features of each of the various embodiments described above may be combined with features of other described embodiments as appropriate in order to provide a multiplicity of feature combinations in associated new embodiments. Furthermore, while the foregoing describes a number of separate embodiments, what has been described herein is merely illustrative of the application of the principles of the present invention. Additionally, although particular methods herein may be illustrated and/or described as being performed in a specific order, the ordering is highly variable within ordinary skill to achieve methods, systems, devices, and software according to the present disclosure. Accordingly, this description is meant to be taken only by way of example, and not to otherwise limit the scope of this invention.

(98) Exemplary embodiments have been disclosed above and illustrated in the accompanying drawings. It will be understood by those skilled in the art that various changes, omissions and

additions may be made to that which is specifically disclosed herein without departing from the spirit and scope of the present invention.

Claims

1. A programmable pill dispensing device, wherein the device comprises: a housing comprising: a channel having a first section and a second section; a storage cavity connected to the first section of the channel, wherein the storage cavity is configured to store a plurality of pills; a photoelectric sensor configured to detect whether a number of pills in the channel is below a threshold; an indicator configured to indicate when the number of pills in the channel is below the threshold as detected by the photoelectric sensor; and a tray connected to the second section of the channel further comprising a pressure sensor configured to detect at least a dispensed pill; a rotational dispensing element, wherein the dispensing element is configured to dispense at least a pill of the plurality of pills, wherein the rotational dispensing element comprises: a body capable of rotating about a central axis; and a structure disposed on the body, wherein the structure is configured to engage with the at least a pill of the plurality of pills, wherein the structure comprises a thread extending from the body wherein the thread comprises: a height, wherein the height is greater than a height of the at least a pill of the plurality of pills; a plurality of roots; and a plurality of flanks, wherein: each root of the plurality of roots are disposed in spaces between adjacent flanks of the plurality of flanks; and each space comprises a width, wherein the width captures the at least a pill of the plurality of pills, such that the at least a pill abuts the adjacent flanks of the plurality of flanks, wherein upon an actuation, the thread permits at least a pill of the plurality of pills to traverse through the channel from the first section to the second section in a quarter revolution of the body and into the connected tray for retrieval; a controller, wherein the controller is configured to: actuate the rotational dispensing element as a function of a current time; and wherein, upon an actuation, the rotational dispensing element is configured to move at least a pill of the plurality of pills from the first section of the channel to the second section of the channel and further into the tray in accordance with a rotation of the body; and a communication component, wherein the communication component is configured to communicate an alert to a remote device when a dispensed pill is not retrieved from the tray within a retrieval time, wherein the retrieval time is determined by the controller implementing a machine-learning model trained with training data correlating a plurality of elements of user data and prescription information to plurality of threshold values.

2. The device of claim 1, wherein the central axis is substantially horizontal.

3. The device of claim 1, wherein actuating the rotational dispensing element comprises: generating a predetermined dispensing time; establishing a security parameter; and actuating the rotational dispensing element when presented with the security parameter at the predetermined dispensing time.

4. The device of claim 3, wherein generating the predetermined dispensing time further comprises: training a machine-learning model using training examples, wherein the training examples comprises a plurality of past retrievals as input correlated to a plurality of retrieval times as output; and generating the predetermined dispensing time using the trained machine-learning model.

5. The device of claim 3, wherein generating the predetermined dispensing time further comprises: determining the predetermined dispensing time to be when a quantity of the plurality of pills is detected, by the at least a sensor, to be less than a predetermined quantity threshold.

6. The device of claim 1, wherein the structure comprises: an indentation having a depth, wherein the indentation is sized so that at least a pill of the plurality of pills can be partially seated within the depth.

7. The device of claim 6, wherein upon the actuation, the indentation moves the at least a pill of the plurality of pills from the first section of the channel to the second section of the channel and

further into the connected tray in a half revolution of the body.

8. The device of claim 1, wherein the controller is further configured to command the device to enter a fail-safe mode when a loss of power is detected.

9. The device of claim 1, further comprising: a security component, wherein the security component is configured to prevent undesired access to the storage cavity and the tray.

10. The device of claim 9, wherein the security component comprises: a fingerprint reader configured to scan a finger of a user to allow access to the storage cavity and the tray.

11. The device of claim 9, wherein the security component comprises: a security sensor, wherein the security sensor is configured to: detect an unauthorized access to the storage cavity and the tray; and transmit a command configuring the device to transit from an unsecured mode to a secured mode.

12. The device of claim 5, further comprising: a backup power source, wherein the controller is further configured to switch a power source of the device to the backup power source when a loss of power is detected.

13. A method of automatedly dispensing a pill using a programmable pill dispensing device, the method comprising: storing a plurality of pills in a storage cavity contained within a housing of a programmable pill dispensing device; indicating, using an indicator, when a number of pills in a channel is below a threshold as detected by a photoelectric sensor; alerting a user to input a security parameter; and actuating a rotational dispensing element of the device using a controller, wherein the rotational dispensing element comprises: a body capable of rotating about a central axis; and a structure disposed on the body, wherein the structure is configured to engage with at least a pill of the plurality of pills, wherein the structure comprises a thread extending from the body wherein the thread comprises: a height, wherein the height is greater than a height of the at least a pill of the plurality of pills; a plurality of roots; and a plurality of flanks, wherein: each root of the plurality of roots are disposed in spaces between adjacent flanks of the plurality of flanks; and each space comprises a width, wherein the width captures the at least a pill of the plurality of pills, such that the at least a pill abuts the adjacent flanks of the plurality of flanks, wherein upon an actuation, the thread permits at least a pill of the plurality of pills to traverse through the channel from a first section to a second section in a quarter revolution of the body and into a connected tray for retrieval; and moving, upon actuation of the rotational dispensing element, the at least a pill of the plurality of pills from a first section of the channel to a second section of the channel, wherein the first section is connected to the storage cavity and the second section is connected to a tray of the device; alerting the user using a pressure sensor communicatively connected to an alarm transducer that the at least a pill is dispensed and ready for retrieval by the user; and altering, using a communication component, to a remote device when a dispensed pill is not retrieved from the tray within a retrieval time, wherein the retrieval time is determined by the controller implementing a machine-learning model trained with training data correlating a plurality of elements of user data and prescription information to plurality of threshold values.

14. The device of claim 1, wherein the device further comprises a camera configured to capture an image as a function of motion in front of the device while the at least a pill is in the tray; wherein the device is configured to transmit the image to a caregiver.

15. The device of claim 1, wherein: the housing further comprises a pressure sensor configured to detect whether a number of pills in the storage cavity is low; and the indicator is further configured to indicate when the number of pills in the storage cavity is low as detected by the pressure sensor.
